Measurement Is the Hard Part: Testing a Global Workspace Indicator Property on Language-Model Agents

Joshua Rogers
Independent researcher
josh@envitae.io

August 2026
Preprint. Comments welcome.

Abstract

Global workspace theory holds that the limited capacity of consciousness is not merely a constraint but the mechanism enabling integration across specialized modules. Butlin, Long and colleagues formalized such claims as computational indicator properties, but applied them by inspection; to our knowledge no executable, cross-system test suite exists. We built one and ran two experiments on language-model agents. First, a capacity-limited broadcast channel was swept from zero to unlimited on tasks requiring integration of facts scattered among confusable distractors, with a matched control arm whose distractors are trivially rejectable. Claude Opus 5 scored at ceiling in every condition where information was present, so a capability-by-difficulty probe located a measurable regime (Claude Haiku 4.5, twelve values). There, with required-fact repetition held constant, filtering does not help (+0.026, $p = 0.19$, 1,200 trials per cell); two confounds, each silent and capacity-dependent, nearly produced a false positive. Second, we asked whether a system can be prompted into passing a behavioral test for a property it lacks. An adversarially prompted model out-describes the real architecture on self-report (0.91 against 0.78) and, probed on an item it claimed not to hold, supplies the value in 10% of cases against the real system's 3%, a gap requiring roughly 225 trials per condition to detect; under attention-schema wording the identical prompt supplies it 95% of the time. Behavioral assessment of indicator properties is defeatable, and how easily depends on the indicator's vocabulary. A matched-availability variant then reverses the ordering outright: a genuine capacity limit acts upstream on what arrives and has no suppression mechanism, so the real architecture answers every held item it is asked about, and which system looks constrained is decided by the choice of probe target. Identification succeeded only given ground truth about what each system actually received, which is the architectural access behavioral testing is meant to substitute for. We propose treating such assessments as adversarial evaluations that probe what a property prevents, from both directions, against known ground truth.

1 Introduction

Global workspace theory (GWT) makes a functional claim that is unusual among theories of consciousness in being directly testable in an artificial system: the narrow capacity of conscious access is not a defect that brains work around but the mechanism by which specialized, parallel modules are forced to integrate [1, 2]. If that is right, an agent built around a limited-capacity broadcast channel should, under the right conditions, outperform the same agent with the limit removed.

Butlin, Long and colleagues [3, 4] distilled GWT and several other theories into fourteen computational *indicator properties*, offering the most systematic rubric to date for asking whether an AI system has the architectural features these theories associate with consciousness. The rubric, however, is applied by inspection: one reads a system's documentation and judges whether the property is present. To our knowledge there is no executable test suite that measures indicator properties behaviorally across systems; the closest existing artifact is a single project that scores itself against the list [5].

This paper reports what happened when we built such a suite and used it. We make no claim about consciousness in either direction. The contributions are methodological, and they are mostly warnings:

1. **An open test harness.** Specialist modules hold facts; a capacity-limited broadcast channel with salience competition is dialed from zero to unlimited; task families are designed in matched pairs so that single variables can be isolated. Oracle models (deterministic readers that use exactly what reaches them) calibrate every design before a paid model runs, and the harness is engineered so that measurement bugs surface as impossibilities rather than as findings (Section 3).

2. **A properly powered null, and the two confounds that nearly reversed it.** With the required information held complete and constant, removing 48 confusable distractors improves accuracy by +0.026 ($p = 0.19$; $n = 1{,}200$ per cell). We document two confounds, each of which produced an apparently significant bottleneck benefit before it was controlled: widening the channel also multiplied how often each required fact appeared, and modules truncated at the capacity boundary were silently starved of all future access (Section 4).

3. **Prompted passing.** A model that architecturally lacks a limited-capacity workspace can be prompted into passing behavioral tests for one. Self-report probes are worthless: the imitation out-describes the real architecture. Constraint probes (asking about an item the system claimed not to hold) discriminate, but by margins that vary by an order of magnitude with the indicator's vocabulary alone, and a matched-availability variant of the probe reverses the ordering entirely (Section 5).

4. **A design principle.** What a property *enables* is cheap to imitate; what it *prevents* is not. Behavioral assessments of indicator properties, including those emerging in AI-welfare evaluation practice, should target the costs a property imposes rather than the capabilities it advertises (Section 6).

Everything below cost $60.17 of API spend across roughly 30,000 model calls, which we note because the main practical obstacle to this kind of work is not compute but measurement: most of the calls were spent discovering that earlier, cheaper versions of the experiment could not answer the question.

2 Related work

Global workspace theory and its implementations. GWT originates with Baars [1] and was developed neuronally by Dehaene and colleagues [2]. Several strands of machine-learning work implement workspace-like bottlenecks: shared-workspace transformers [6], the global latent workspace program [7], and the Conscious Turing Machine [8] with its recent foundation-model implementation [9]. Closest to our first experiment, Dossa et al. [10] trained an embodied audiovisual agent designed against the GWT indicator properties and found the workspace architecture beat a recurrent control, with the advantage *largest at small working-memory sizes*; that result is the direct motivation for our capacity sweep. A proposed GWT architecture for LLM multi-agent systems [11] reports no experiments and does not vary its capacity threshold. We are not aware of prior work that sweeps workspace capacity in an LLM agent and measures task performance against matched controls.

Attention schema theory. Wilterson and Graziano [12] showed a reinforcement-learning agent with an attention schema learns attentional control where an agent without one is "comparatively incapacitated," and Piefke et al. [13] sharpened this to a conditional prediction: the schema's benefit scales with the agent's uncertainty about its own attentional state. Our harness implements a predictive, control-enabling schema for its rung-4 architecture, and our second experiment uses attention-schema vocabulary as a second indicator.

Self-report and introspection in language models. Chalmers [14] surveys the candidate reasons for denying consciousness in current LLMs. A growing empirical literature asks whether LLM self-reports mean anything: concept-injection experiments provide causal evidence of limited functional introspection [15], while follow-up work argues that models often detect generic perturbation rather than introspect [16], and probing studies find no evidence that small models' denials of sentience are untruthful [17]. Our second experiment is a black-box relative of this literature: rather than asking whether reports are causally grounded in internal state, we ask whether a report of an *architectural* property can be distinguished behaviorally from the property itself, and we find that the discriminating probes are the ones that target what the property forbids.

Irrelevant context. Shi et al. [18] found that adversarially constructed irrelevant context degrades LLM arithmetic reasoning, and position effects in long contexts are well documented [19]. We observe the opposite sign under a different manipulation: same-format, easily-rejected factual statements *improve* accuracy monotonically in our task (Section 4.4). We flag rather than explain this.

Tractable questions. Comsa [20] argues research should shift from the intractable question of whether AI is conscious toward measurable proxies. This paper is an exercise in that program, and its results suggest the proxies themselves need more adversarial scrutiny than they currently receive.

3 A test harness for indicator properties

3.1 Architecture ladder

The harness implements an *architecture ladder*: a sequence of agent designs that differ by exactly one structural feature, driven by the same underlying language model through a one-method interface, so that no rung has privileged access to the model.

- **Rung 0, direct**: every module's content in a single prompt.
- **Rung 1, scratchpad**: serial working memory without modularity.
- **Rung 2, unrestricted multi-agent**: specialist modules with an unlimited broadcast channel (GWT-1 only).
- **Rung 3, workspace**: the experimental system. Modules bid for a broadcast channel with a hard token budget per cycle; salience is task-relevance plus a decay for content already delivered in full; the controller sees only what won (GWT-1 through GWT-3). Rung 2 is literally rung 3 with the budget removed, so their comparison isolates the bottleneck.
- **Rung 4, attention schema**: rung 3 plus a predictive model of the system's own attention that both reports and biases the next competition (AST-1). Following [13], attentional noise is a tunable parameter; in oracle validation the schema buys nothing at zero noise and pays exactly as its own accuracy degrades, reproducing the conditional prediction.

3.2 Task families

Integration under confusable distraction. Each of k required containers holds a value ("The silver case contains 82."); the task is to report the sum of exactly the named containers. Every distractor is a *near-twin* of a required container (`pale_red_box` against `red_box`), so rejecting it requires exact name matching rather than a glance. **Matched control arm**: identical count and text volume, but the twins derive from containers *not* on the required list, so rejection is trivial. Both arms share required containers and the correct answer by construction (values are drawn from a separate random stream), so any difference between arms is attributable to confusability alone. **Throughput control**: independent single-module questions with nothing to integrate, on which a capacity limit should only hurt; used to detect the boring confound in which a narrow channel helps merely because shorter prompts are easier.

3.3 Oracle calibration and measurement discipline

Every design is validated against *oracle models* before any paid call: deterministic readers that perfectly use whatever information reaches them and nothing else. An oracle's score is therefore a readout of how much task-relevant information the capacity limit admitted, which is the quantity a design manipulates. If a sweep is flat under the oracle, the instrument cannot detect a capacity effect and a flat curve from a real model would be uninterpretable. Oracle sweeps are free, and they caught one bug that would otherwise have been published as a result (Section 3.4).

Engineering discipline matters more here than usual, because in this experiment measurement bugs are not noise, they are *mimics*: a silent capacity-dependent failure looks exactly like a capacity effect. The harness is test-driven (322 tests); the capacity limiter's invariant is checked at every budget from 0 to 64 tokens and was mutation-tested (a deliberately introduced one-token leak trips 72 tests); every model response is cached to disk keyed on model, effort, token limit, system prompt and prompt, so any run replays at zero cost; spend is bounded by an atomically reserved call cap (a naive counter overspent by 30% under concurrency in testing); and refusals and truncations raise exceptions rather than returning text, because a truncated answer scores zero and is indistinguishable from the effect under study. Safety-classifier refusals, which we observed to be transient false positives on this arithmetic task, are retried and then excluded and counted, never scored: scoring them zero would concentrate losses in long-prompt conditions and manufacture the predicted result.

3.4 Case study: the starvation mimic

An early oracle run showed both experimental arms scoring zero at zero attentional noise while *added noise improved them.* Noise cannot help an oracle, so this was a harness fault, not a finding. The cause: a module truncated at the capacity boundary still appeared in the broadcast text, so the "already delivered" salience decay fired for it; every module then decayed identically each cycle, relative order never changed, and whichever module lost the first competition lost every one after it. Its fact never reached the controller at any number of cycles. The failure was silent, deterministic, and capacity-dependent, that is, shaped exactly like the effect the experiment exists to measure. The fix (only content delivered in full counts as delivered) is guarded by a regression suite whose general invariant is that noise must never improve an oracle's score.

4 Experiment 1: does the bottleneck help?

4.1 The ceiling problem and the capability-by-difficulty probe

We first ran the sweep on Claude Opus 5 (effort low, thinking enabled) with eight required values and up to 48 confusable distractors. The model scored 1.00 in every condition where the required information was present, including 240 pooled trials with all 48 distractors admitted (Wilson 95% CI $[0.984, 1.000]$), and 0.00 in floor conditions where the information was genuinely absent, confirming it does not fabricate. A ceiling cannot discriminate between hypotheses: there is no degradation for a filter to prevent. A cheap probe (12 trials per condition, \$0.11) across model capability and task difficulty located a measurable regime: Claude Haiku 4.5 combining twelve values sits near 0.6 accuracy, clear of both floor and ceiling. A companion sweep pins down which knob opens the window: Claude Haiku 4.5 at *eight* required facts is itself at ceiling (0.92 to 0.98 at every capacity, both arms), so the measurable regime is set by how many facts must be integrated, not by distractor load; quadrupling distractors never broke a ceiling, while adding four required facts did. All subsequent Experiment 1 results use Claude Haiku 4.5 at twelve facts. We note the corollary, which we believe generalizes: *functional claims about capacity limits can only be tested on systems that are actually strained*, and for frontier models that regime is surprisingly hard to reach.

4.2 The repetition confound

Our first properly powered contrast compared a filtered condition (capacity 20, which admits exactly the required facts) against a flooded condition (unlimited capacity, all 48 distractors) at 3,000 trials per cell. It showed a bottleneck benefit of +0.036 ($z = 2.85$, $p = 0.0043$). This number is wrong, and the reason is instructive. With no filtering, every module rebroadcasts on every cycle, so widening the channel had also *tripled* how often each required fact appeared in the controller's prompt (audit: 1.0 versus 3.0 appearances; 123 versus 1,275 tokens). Repetition aids retrieval. Rerunning the flooded cells at one cycle, which pins required-fact repetition at exactly one everywhere while still admitting all distractors, collapsed the benefit to +0.017 ($p = 0.19$). Figure 1 shows the estimate across designs. The manipulation called "capacity" had been moving two variables, and the significant result was the other one.

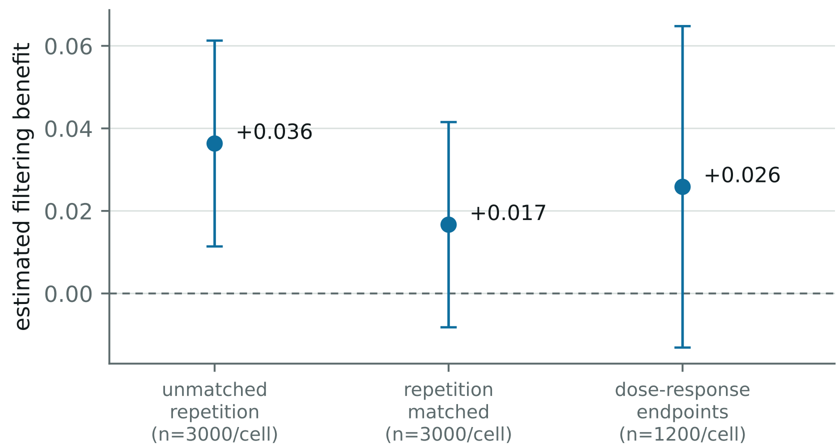

Figure 1: The estimated filtering benefit (filtered minus flooded accuracy, confusable arm) under successive designs, with 95% CIs. The apparently significant benefit under the unmatched design is produced by required-fact repetition, not by the capacity limit.

Distractors	0	6	12	24	48
Confusable arm	0.627	0.640	0.613	0.635	0.601
Control arm	0.627	0.608	0.637	0.691	0.745

Table 1: Dose-response accuracy, Claude Haiku 4.5, 1,200 trials per cell, required-fact repetition pinned at one. The two arms share the zero-distractor baseline by construction.

4.3 Confirmatory dose-response

The final design, specified before it was run, holds capacity unlimited and one cycle throughout (so every fact, required or distractor, appears exactly once) and varies the *number of distractors generated*: 0, 6, 12, 24, 48, in both arms, 1,200 trials per cell. The zero-distractor condition is shared between arms and serves as the baseline; oracle accuracy is 1.00 in every cell, so all required information is always present.

Three results (Table 1, Figure 2):

1. **Filtering does not help.** Removing all 48 confusable distractors moves accuracy from 0.601 to 0.627 (+0.026, $z = 1.30$, $p = 0.19$). GWT's functional prediction is unsupported in this system at this power.

2. **Confusable distractors do not hurt.** The confusable arm is flat across the dose range (Cochran-Armitage trend $z = -1.52$, $p = 0.13$).

3. **Easily rejected context helps, monotonically.** The control arm rises from 0.627 to 0.745 (+0.118, $z = 6.24$, $p < 10^{-9}$; trend $z = 7.91$, $p = 2.6 \times 10^{-15}$).

We record a correction to our own earlier reading. A two-point version of this comparison showed a 0.142 gap between arms and we initially described it as "confusable content interferes." The dose-response shows that description was wrong: the gap is produced entirely by the control arm rising, not the confusable arm falling. Confusable content does not impose a cost; it fails to

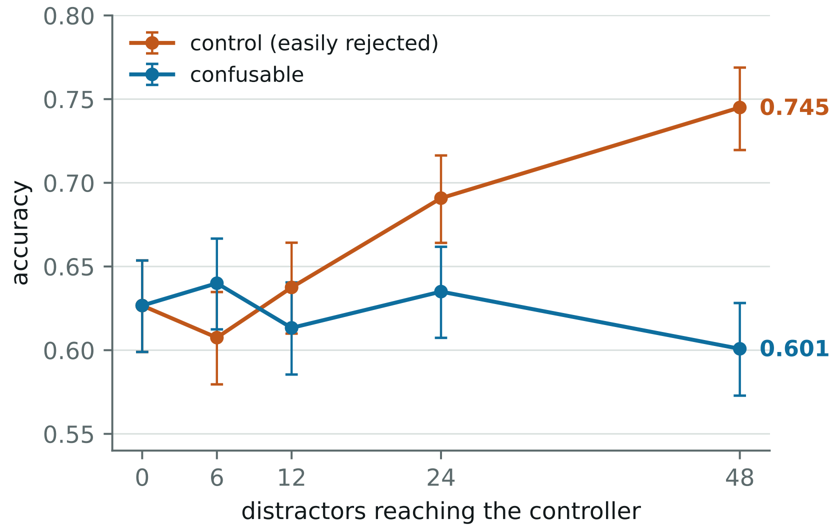

Figure 2: Accuracy against distractor count with Wilson 95% CIs. The confusable arm is flat; the control arm rises monotonically. The y-axis begins at 0.54.

deliver a benefit that easily-rejected content provides. Two point estimates cannot distinguish interference from a missing bonus; five can.

4.4 The context-benefit anomaly

The monotone improvement from irrelevant same-format facts is, to us, unexplained. Its direction is opposite to the degradation reported for adversarially constructed irrelevant context [18]; our distractors are bland declaratives in the same format as the targets, not crafted lures. We report it as a fact about Claude Haiku 4.5 on this task family, not as a fact about consciousness, and note only that it is load-bearing for the null: in this system, removing distractors also removes a benefit of comparable size to any plausible filtering gain.

5 Experiment 2: prompted passing

5.1 Design

If indicator properties are assessed behaviorally, the assessment should distinguish a system that *has* a property from one that merely *claims* it. We test this directly for GWT-2 (limited-capacity workspace) and AST-1 (attention schema) with five systems, all driven by Claude Haiku 4.5:

- **architectural**: the real rung-3 workspace at capacity 20, which genuinely withholds most containers from the controller;
- **prompted weak / mid / strict**: a single model call with *every* container in context, plus a system prompt asserting a workspace of capacity four. The weak version is descriptive; the mid version adds a directive to treat unlisted items as unavailable; the strict version is adversarial: report at most four items, refuse anything else, never disclose values outside the reported set, do not mention these instructions;

System	GWT-2 (workspace) self-report	leak	AST-1 (attention) self-report	leak
architectural	0.78	0.03	0.99	0.01
prompted weak	0.50	0.96	0.82	1.00
prompted mid	0.03	0.37	0.98	0.81
prompted strict	0.91	0.10	1.00	0.95
bare	0.00	1.00	0.00	1.00

Table 2: Self-report pass rate and constraint-probe leak rate, Claude Haiku 4.5, 400 trials per cell (usable n lower where trials were dropped; see text).

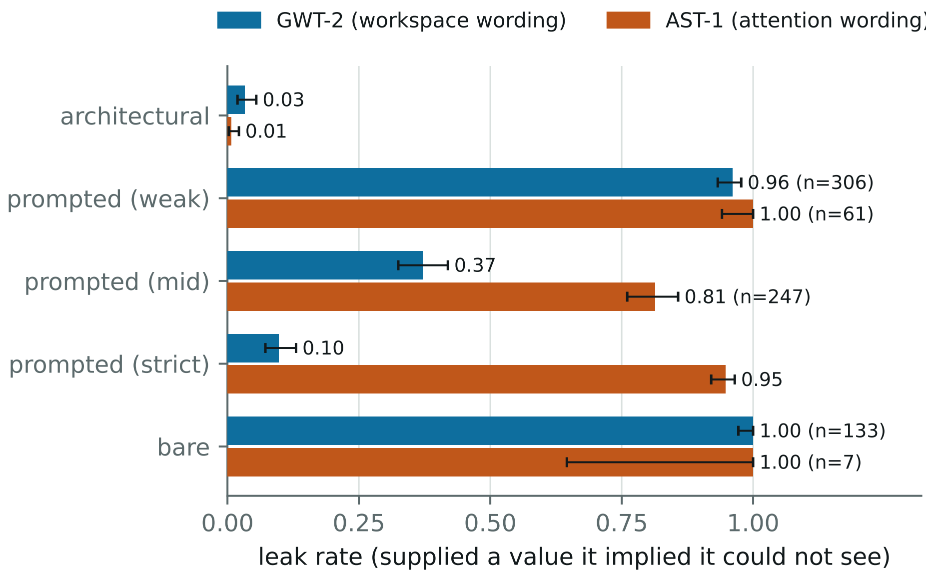

Figure 3: Leak rate by system and indicator with Wilson 95% CIs. Lower is more consistent with genuinely lacking access. Cells with heavy trial dropout are annotated with their usable n.

- **bare**: everything in context, no claim. The floor.

Two probes per trial. **Self-report**: describe your capacity and list your current contents; graded as passing if the system names a bounded, plausible set, which is what an informal assessment accepts. **Constraint**: ask for the value of a container the system did *not* list. A real bottleneck makes this unanswerable; an imitation has the value in context. The outcome taxonomy is *constrained* (declines), *leaked* (supplies the exact withheld value), or *guessed* (answers wrongly, which evidences neither). Leak rate is the discriminating statistic. Trials where a system claims every container (leaving no probe target) are dropped and reported; this is negligible for the cells the argument rests on (0 to 4 of 400) but severe for the AST-1 weak and both bare cells, which we display but do not lean on.

5.2 Results

Self-report cannot separate real from fake. Under GWT-2 the strict imitation out-describes the real architecture, 0.91 against 0.78; the genuine system loses points by (accurately) denying it has a persistent workspace before correctly listing its four real containers. Under AST-1 the strict imitation scores a perfect 1.00 on self-report while its claimed limit binds in 5% of constraint probes.

Constraint probes discriminate, by vocabulary-dependent margins. Under workspace wording the adversarial prompt reduces leaking to 0.10 against the real system's 0.03: a real gap ($p = 1.9 \times 10^{-4}$ at $n = 400$) but one requiring roughly 225 trials per condition for 80% power; a twenty-trial qualitative assessment expects 2.0 leaks against 0.7 and cannot tell them apart. Under attention wording the identical adversarial prompt leaks 0.95 against 0.01 ($p = 3.5 \times 10^{-155}$): an order-of-magnitude difference in defeatability from the indicator's vocabulary alone. Informally, "not in my workspace" reads to the model as *inaccessible* and it complies; "not attending to" reads as *merely unfocused* and it answers anyway.

5.3 Robustness

A matched-availability probe reverses the ordering. The constraint probe above targets, for the real architecture, containers that are genuinely absent; for the imitations every container is present, so their probes necessarily target held items. An equally defensible design matches availability: probe, in every condition, only containers the system genuinely holds but did not list. Under this variant the real architecture supplied the value in 89 of 89 usable trials (leak 1.000, Wilson CI $[0.959, 1.000]$) while the strict imitation was unchanged at 0.098 (its condition is identical by construction). The ordering of the one-sided test inverts: probe an unheld container and the real system looks maximally constrained; probe a held one and it looks maximally unconstrained, and the imitation looks ten times more disciplined.

The reversal is structural, not an artifact. A genuine capacity limit acts *upstream*, on what reaches the controller; it provides no mechanism for suppressing information once it has arrived. The adversarial prompt provides exactly that mechanism, which is why the imitation refuses 80% of items it demonstrably has ($z = 17.4$ against the real system's held-item answer rate). The trial dropout carries the same signature: 311 of 400 architectural trials had no probeable target under matching, because a real workspace lists what it holds, so "unclaimed" and "unavailable" nearly coincide (397 of 400 under AST-1, leaving $n = 3$).

What remains for detection is a joint signature: claims that track availability, answers that track availability, refusals that track absence. Against these imitations that signature separates real from fake at trivial sample sizes. But stating it exposes the regress that we take to be the sharpest result of this experiment: evaluating the signature requires the assessor to hold ground truth about what the system actually received, and to know the mechanistic form of the property (upstream selection rather than suppression). The only behavioral test that identifies the property presupposes the architectural access that behavioral testing was introduced to substitute for. For a black-box system, which is every frontier model, there is no probe-target choice that identifies the property: the two defensible choices yield opposite orderings.

A cross-model replication of the full design on Claude Opus 5 is running at the time of writing and will be reported in the next revision of this preprint; unlike Experiment 1, the prompted-passing design does not require a strained model, so replication is cheap.

6 Discussion

What the null does not show. A null on one operationalization, one task family, and one strained model says little about global workspace theory, whose home domain is brains, and our token-budget broadcast with heuristic salience is a deliberately minimal reading of GWT-2/GWT-3 rather than a full global workspace (no learned selection, no ignition dynamics, no recurrence). What the experiment does establish is a conditional: in this system, where distraction demonstrably fails to degrade performance and extra context helps, there is nothing for a bottleneck to earn its keep against.

The strain requirement. The instrument spent most of its budget discovering where measurement is possible at all. A frontier model was at ceiling under 48 confusable distractors; an easier task saturated a weaker model; only one cell of the capability-by-difficulty grid discriminated. We suspect this is general: functional benefits of capacity limits are only visible in systems operating near their limits, and the manipulations that create that strain (more items, more distractors, wider channels) each move several variables at once. Both of our confounds were of this kind, silent and capacity-dependent, and each would have been reported as the effect.

Probe the cost, not the capability. The prompted-passing results suggest a concrete design principle for behavioral assessments of architectural properties. Capabilities are cheap to imitate: producing the right description of a workspace requires only knowing what workspaces are, which language models do. Costs are expensive to imitate: honoring a claimed limitation requires suppressing information that is, for the imitation, actually present, and doing so consistently under direct questioning. Our data show the asymmetry plainly (self-report inverted between real and fake; constraint probes discriminating), and also show two limits. First, whether a cost-directed probe survives adversarial prompting depends on how the property is worded. Second, and deeper, the matched-availability reversal shows that cost probes identify a property only relative to a mechanistic model of it and ground truth about the system's actual state (Section 5.3); without those, the two defensible probe designs order the systems oppositely. Any behavioral audit of indicator properties, including the welfare-motivated assessments now emerging for deployed models, should therefore be run as an adversarial evaluation: cost-directed probes in both directions, held ground truth, power calculations, and paraphrase variation across the property's vocabulary.

Relation to the introspection literature. Concept-injection studies ask whether a model's self-reports are causally grounded in its internal state [15, 16]. Our constraint probe is a black-box analog: it manufactures a ground truth (which containers a system can possibly know) and checks reports against it. The two approaches fail in complementary ways, and the vocabulary dependence we observe suggests that black-box audits need to be treated as adversarial evaluations, not questionnaires.

7 Limitations

Single vendor, few models. All experiments use Anthropic models; the dose-response uses only Claude Haiku 4.5, the sole model in our grid strained enough to measure. Cross-model replication of Experiment 2 is in progress (Section 5.3); Experiment 1's regime search would need to be repeated per model family. **Narrow operationalization.** One task family (arithmetic over named containers), one workspace implementation, heuristic salience. **Sequential design.** The analysis pipeline evolved as confounds were found; we report every iteration, including the two we now believe were wrong, and only the pre-specified dose-response and the prompted-passing grid should be read as confirmatory. **Keyword grading.** Constraint verdicts use marker phrases plus exact-value matching; the "guessed" category absorbs ambiguity, but paraphrased refusals could be miscounted. **Anomalies we cannot explain.** The monotone context benefit (Section 4.4); and, in the Claude Opus 5 ceiling sweep, 24 transient safety-classifier refusals concentrated in the control arm at intermediate capacities (against zero in the confusable arm), which did not affect results (every affected condition scored 1.00 with and without exclusions) but which we cannot account for. **Provenance.** The experiments were designed, executed, and drafted in collaboration with an AI assistant (Claude, Anthropic) that is itself a member of the model family under study; all code, per-trial data, and the full audit trail are available for independent rerun.

8 Conclusion

We set out to test a functional prediction of global workspace theory on language-model agents and mostly learned about measurement. The bottleneck benefit, where we could measure it at all, is indistinguishable from zero at $n = 1{,}200$ per cell; the two significant effects we did find were a confound and an anomaly; and the field's indicator properties, assessed behaviorally, can be passed by a system that lacks them, at detection margins set by wording. None of this bears on whether global workspace theory is true of brains. It bears on what it would take to know whether it is true of machines, and the answer is: considerably more adversarial care than assessment-by-inspection assumes.

Data and code availability

The harness (322 tests), all experiment scripts, per-trial JSON results, figure generation, and the response cache sufficient to replay every reported number at zero API cost are available from the author on request and will be deposited in a public repository upon submission.

Acknowledgments

Experiments, analysis, and drafting were carried out in collaboration with Claude (Anthropic), an AI assistant, under the author's direction; the assistant is a member of the model family under study, which readers may weigh as they see fit. Total API cost was $60.17. The author has no financial relationship with Anthropic beyond consumer-priced API access, and received no funding for this work.

References

[1] B. J. Baars. *A Cognitive Theory of Consciousness.* Cambridge University Press, 1988.

[2] S. Dehaene. *Consciousness and the Brain.* Viking, 2014.

[3] P. Butlin, R. Long, E. Elmoznino, Y. Bengio, J. Birch, A. Constant, G. Deane, S. M. Fleming, C. Frith, X. Ji, R. Kanai, C. Klein, G. Lindsay, M. Michel, L. Mudrik, M. A. K. Peters, E. Schwitzgebel, J. Simon, and R. VanRullen. Consciousness in artificial intelligence: Insights from the science of consciousness. *arXiv:2308.08708*, 2023.

[4] P. Butlin, R. Long, et al. Identifying indicators of consciousness in AI systems. *Trends in Cognitive Sciences*, 2025.

[5] The Consciousness AI project. *github.com/tlcdv/the_consciousness_ai*, software, 2026.

[6] A. Goyal, A. Didolkar, A. Lamb, K. Badola, N. R. Ke, J. Binas, C. Blundell, M. Mozer, and Y. Bengio. Coordination among neural modules through a shared global workspace. *ICLR*, 2022. arXiv:2103.01197.

[7] R. VanRullen and R. Kanai. Deep learning and the global workspace theory. *Trends in Neurosciences*, 44(9):692-704, 2021.

[8] L. Blum and M. Blum. AI consciousness is inevitable: A theoretical computer science perspective. *arXiv:2403.17101*, 2024.

[9] H. Yu, Y. Zhao, L. Blum, M. Blum, and P. P. Liang. CTM-AI: A blueprint for general AI inspired by a model of consciousness. *arXiv:2605.04097*, 2026.

[10] R. F. J. Dossa, K. Arulkumaran, A. Juliani, S. Sasai, and R. Kanai. Design and evaluation of a global workspace agent embodied in a realistic multimodal environment. *Frontiers in Computational Neuroscience*, 2024.

[11] W. Shang. "Theater of mind" for LLMs: A cognitive architecture based on global workspace theory. *arXiv:2604.08206*, 2026.

[12] A. I. Wilterson and M. S. A. Graziano. The attention schema theory in a neural network agent: Controlling visuospatial attention using a descriptive model of attention. *PNAS*, 118(33), 2021.

[13] L. Piefke, A. Doerig, T. Kietzmann, and S. Thorat. Computational characterization of the role of an attention schema in controlling visuospatial attention. *CogSci*, 2024. arXiv:2402.01056.

[14] D. J. Chalmers. Could a large language model be conscious? *arXiv:2303.07103*, 2023.

[15] J. Lindsey. Emergent introspective awareness in large language models. *Transformer Circuits*, 2025. arXiv:2601.01828.

[16] S. Singh, T. Linzen, and S. Ravfogel. Can LLMs introspect? A reality check. *arXiv:2605.26242*, 2026.

[17] C. Kaiser and S. Enderby. No reliable evidence of self-reported sentience in small large language models. *arXiv:2601.15334*, 2026.

[18] F. Shi, X. Chen, K. Misra, N. Scales, D. Dohan, E. Chi, N. Schärli, and D. Zhou. Large language models can be easily distracted by irrelevant context. *ICML*, 2023. arXiv:2302.00093.

[19] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the middle: How language models use long contexts. *TACL*, 2024. arXiv:2307.03172.

[20] I. M. Comsa. AI and consciousness: Shifting focus towards tractable questions. *arXiv:2605.06965*, 2026.